

SMART MULTI-USER EXPENSE TRACKER AND BUDGET MANAGEMENT SYSTEM

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Internship Project Report

Smart Multi-User Expense Tracker and Budget Management System

Project Report

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Introduction

Managing personal finances is an important part of daily life. Many people find it difficult to track expenses, savings, and budgets manually. This project provides a smart solution for monitoring expenses and managing financial activities efficiently.

The Smart Multi-User Expense Tracker is a web-based application developed using Python and Streamlit. The system allows multiple users to register, log in, add expenses, monitor budgets, analyze savings, and visualize spending through charts and reports.

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Abstract

The Smart Multi-User Expense Tracker and Budget Management System is designed to help users manage their daily financial activities. The application supports user authentication, expense management, budget tracking, savings calculation, and graphical visualization of expenses.

Users can create accounts, securely log in, add expense details, set monthly budgets, and analyze their spending patterns. The system also generates alerts when the budget limit is exceeded and provides downloadable Excel reports.

The project was developed using Python libraries such as Streamlit, pandas, matplotlib, and openpyxl.

Objectives

To develop a multi-user expense tracking system.

To monitor user spending and savings.

To generate charts for expense analysis.

To provide budget alerts and financial summaries.

To export expense reports in Excel format.

Tools and Technologies Used

Features of the Project

User Registration and Login

Multi-user Support

Expense Tracking

Budget Monitoring

Savings Calculation

Expense Visualization using Charts

Budget Warning Alerts

Excel Report Export

Real-Time Financial Summary

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Steps Involved in Building the Project

Step 1: Requirement Analysis

The project requirements were analyzed to understand the features needed for expense management.

Step 2: Designing the System

The structure of the application, user authentication, and dashboard design were planned.

Step 3: User Authentication

A login and registration system was created using CSV files for storing user credentials.

Step 4: Expense Management

Users were allowed to add expenses with categories and amounts.

Step 5: Data Visualization

Charts such as bar graphs and pie charts were generated using matplotlib.

Step 6: Budget and Savings Calculation

The system calculates total spending, remaining budget, and savings automatically.

Step 7: Report Generation

Expense reports were exported into Excel format using openpyxl.

Step 8: Deployment

The project was uploaded to GitHub and deployed using Streamlit Cloud.

System Workflow

1. User registers account.
2. User logs into the application.
3. User adds expense details.
4. System stores user-wise data.
5. Charts and financial summaries are generated.
6. Budget alerts are shown if spending exceeds limits.
7. Reports can be exported to Excel.

Advantages of the Project

Easy expense tracking

User-friendly interface

Real-time financial analysis

Budget monitoring system

Multi-user support

Automated report generation

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Future Enhancements

Mobile application version

Database integration using MySQL/Firebase

AI-based expense prediction

Cloud synchronization

Dark mode support

Notification reminders

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Conclusion

The Smart Multi-User Expense Tracker and Budget Management System successfully helps users manage expenses, monitor budgets, and analyze savings effectively. The project provides an interactive and user-friendly interface with real-time financial insights and visualization features. This system can be further enhanced with advanced technologies for real-world financial management applications.